

Machine Learning

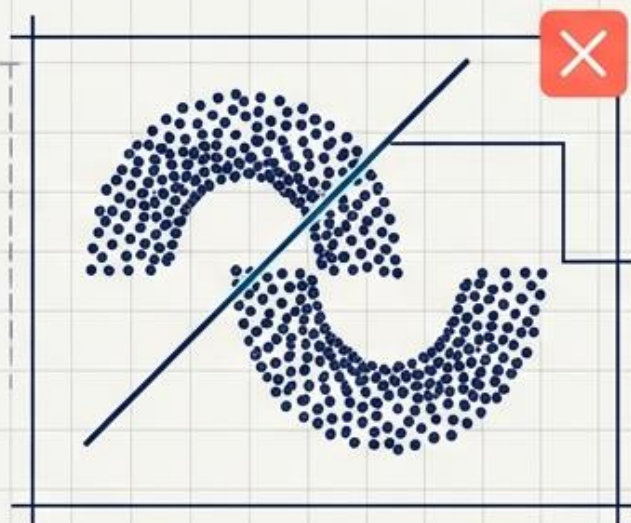
Week 7 Lecture

Dr. Muhammad Ahmed Raza

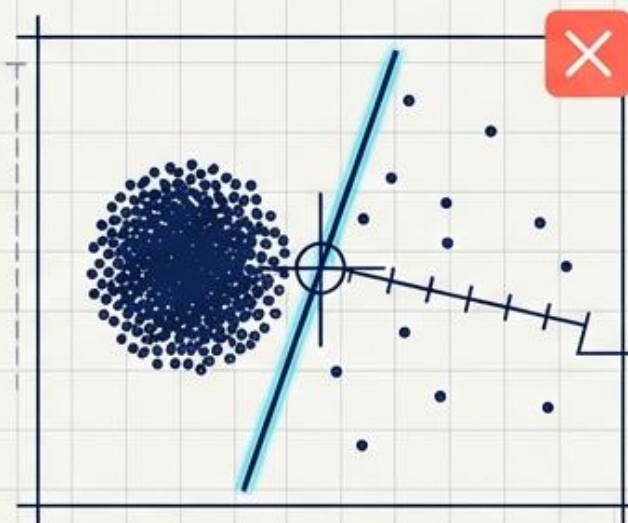
Contents

- Limitations of Partition Algorithms (Flat)
- Intro to Hierarchical Algorithms
- Agglomerative Clustering
 - Single linkage
 - Complete Linkage
 - Discovering k

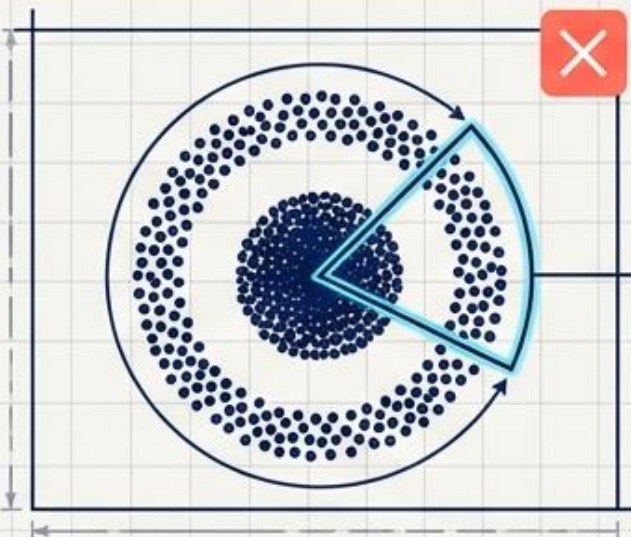
The Limits of Flat Clustering



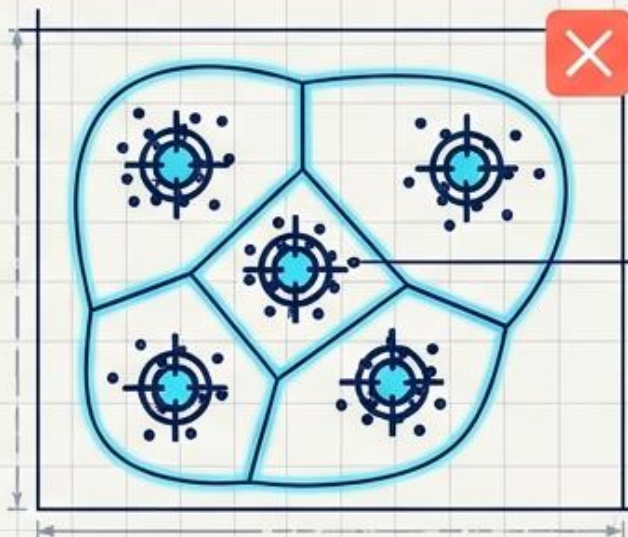
Non-spherical shapes:
Assumes all clusters are perfectly spherical.



Varying cluster densities:
Assumes equal variance across all groups.



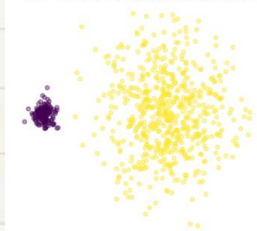
Nested structures:
Fails to recognize clusters entirely contained within others.



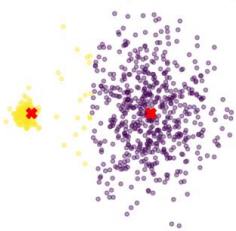
The "k" problem:
Requires a pre-defined k. Real-world data is rarely this neat.

k = ?

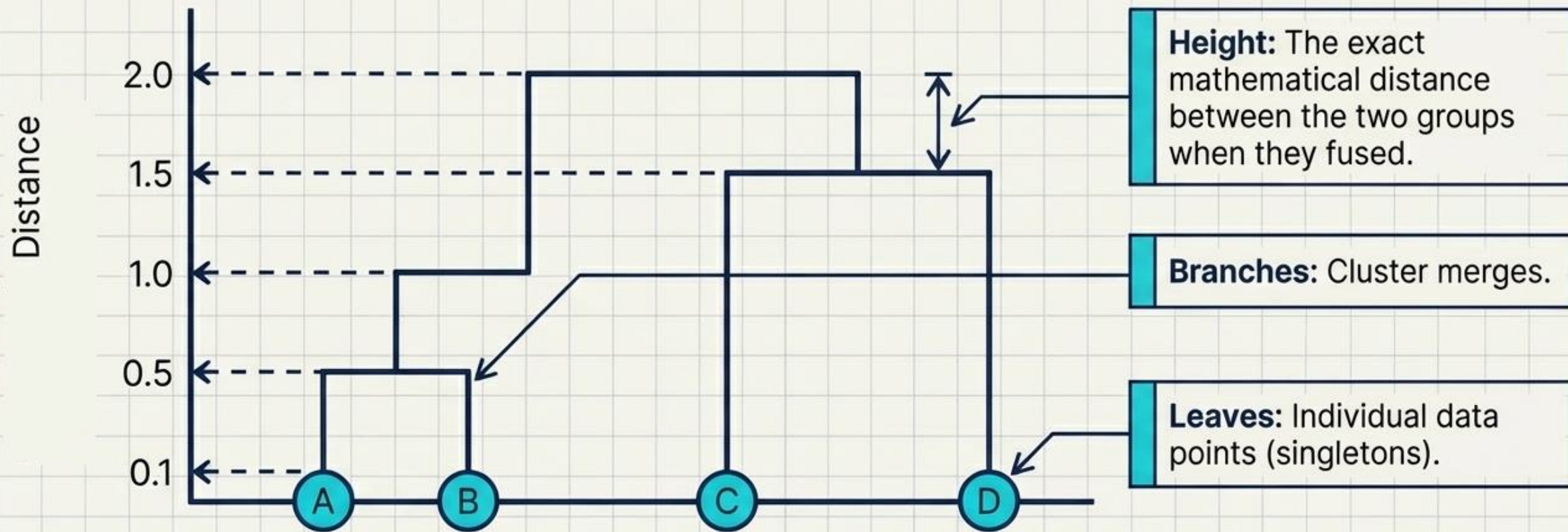
True Clusters (Density/Size Mismatch)



K-means Result (Forced Balance)



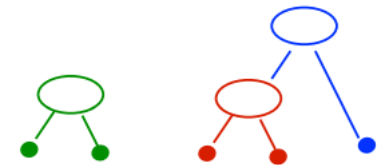
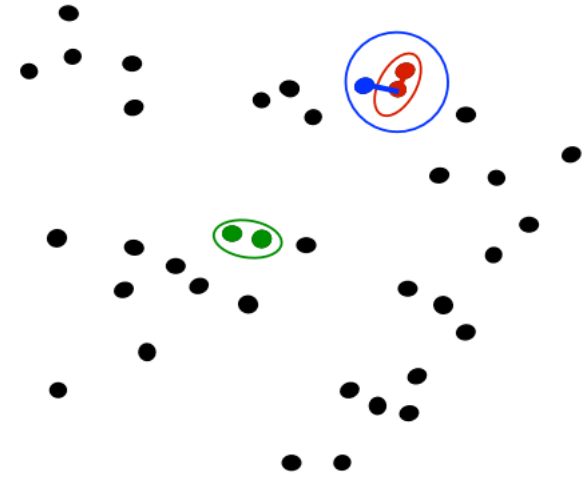
"Anatomy of a Dendrogram" in Space Grotesk



The Golden Rule: You cannot judge similarity by horizontal proximity on the X-axis; similarity is dictated only by the vertical fusion height.

Agglomerative Clustering

- Agglomerative clustering:
 - First merge very similar instances
 - Incrementally build larger clusters out of smaller clusters
- Algorithm:
 - Maintain a set of clusters
 - Initially, each instance in its own cluster
 - Repeat:
 - Pick the two closest clusters
 - Merge them into a new cluster
 - Stop when there's only one cluster left
- Produces not one clustering, but a family of clusterings represented by a dendrogram



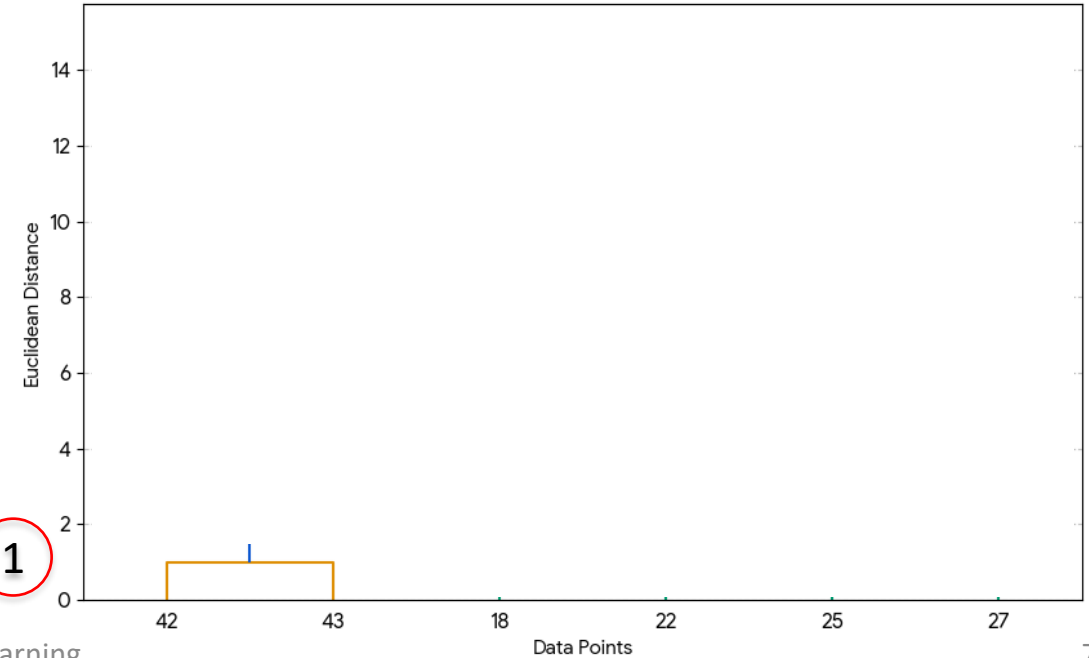
Agglomerative Clustering (Example)

- Consider the following 6 one dimensional data points:
- 18,22,25,42,27,43
- Apply agglomerative clustering algorithm to build a dendrogram
- Merge the clusters using minimum distance and update the proximity matrix accordingly

Step 1

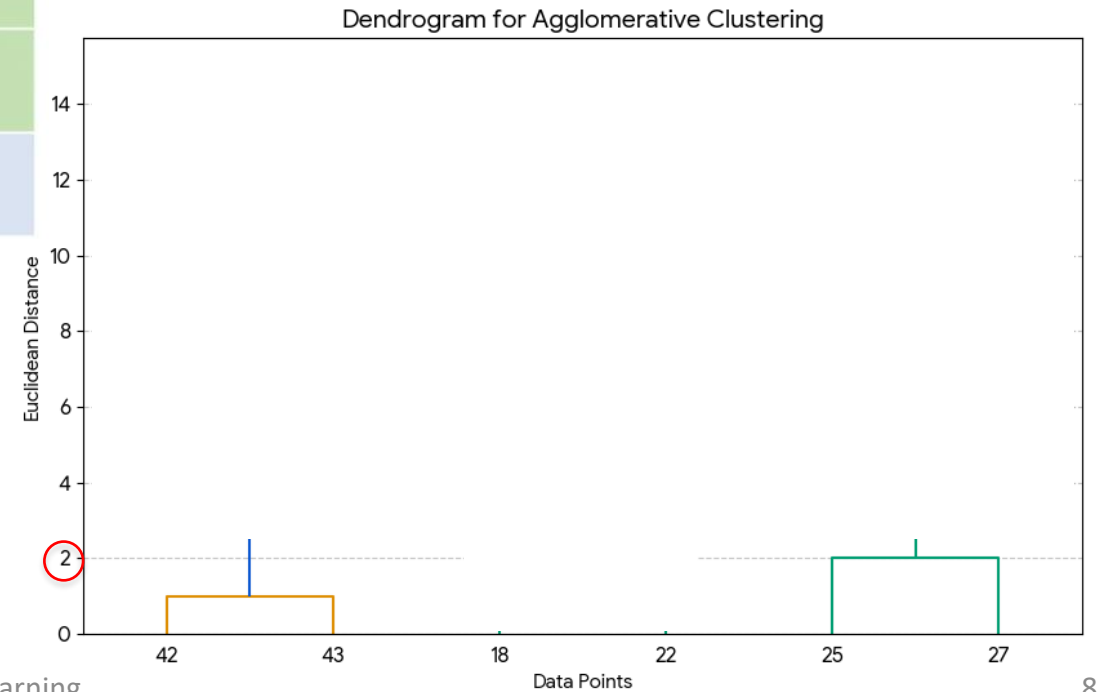
	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

Dendrogram for Agglomerative Clustering



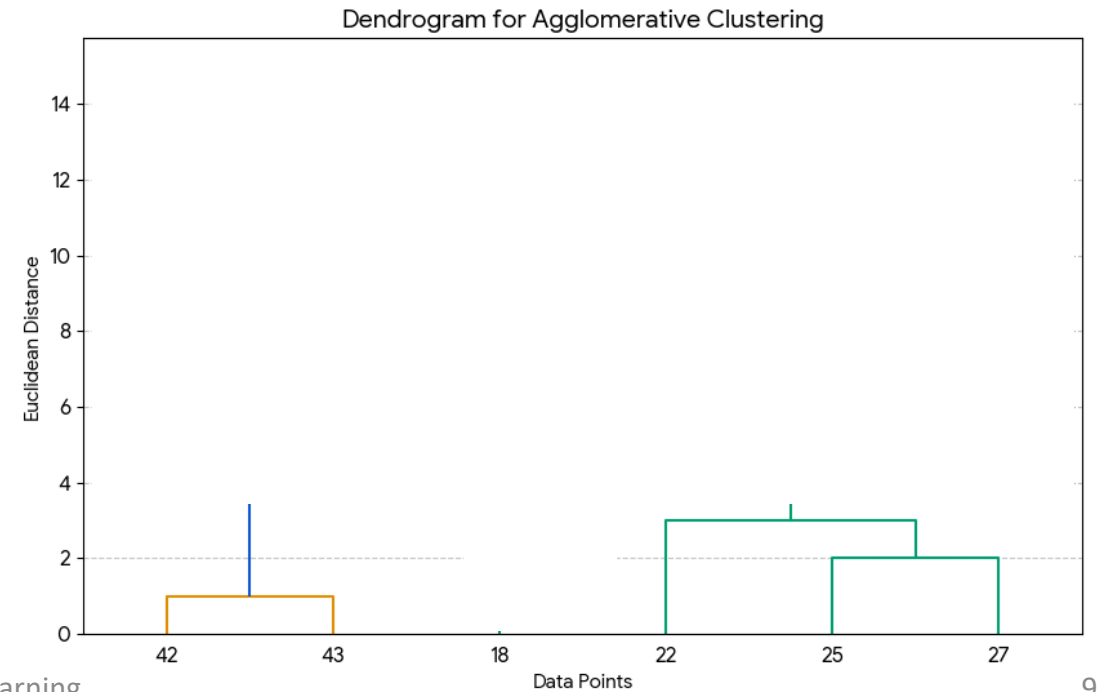
Step 2

	18	22	25	27	42, 43
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	2	17
27	9	5	2	0	15
42, 43	24	20	17	15	0



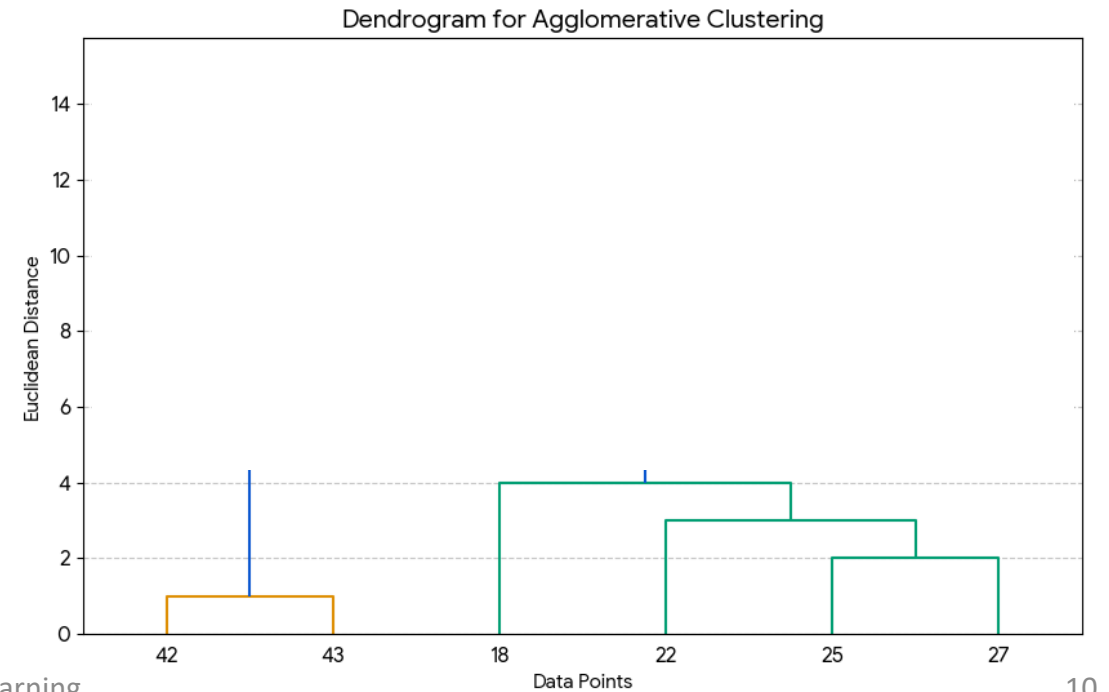
Step 3

	18	22	25, 27	42, 43
18	0	4	7	24
22	4	0	3	20
25, 27	7	3	0	15
42, 43	24	20	15	0



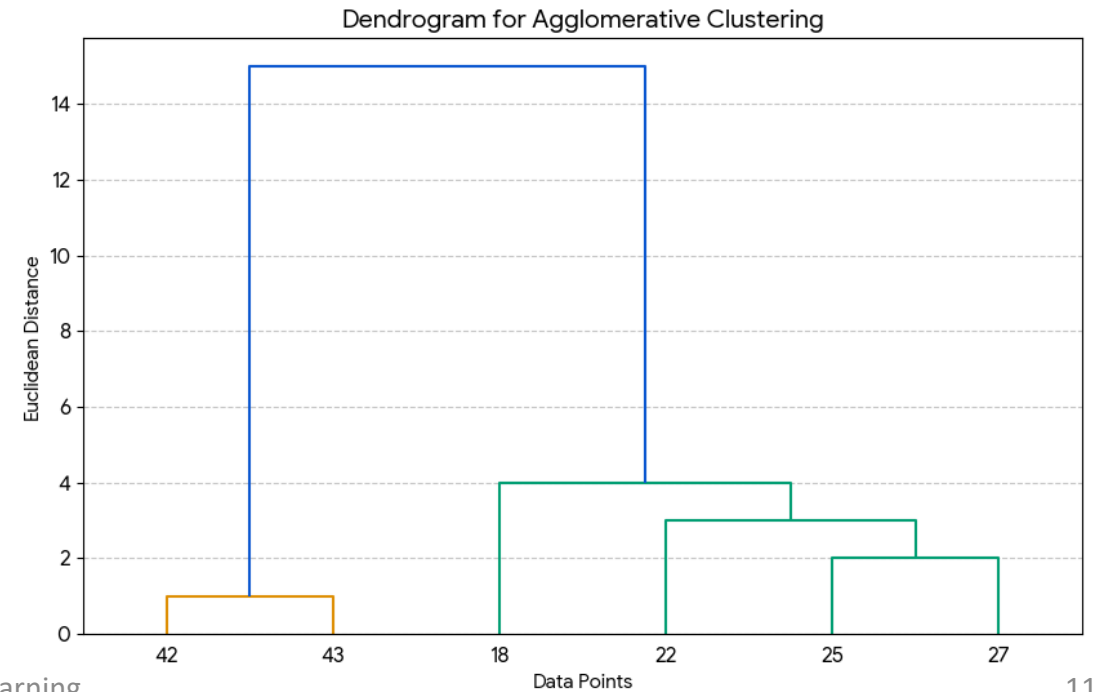
Step 4

	18	22,25,27	42, 43
18	0	4	24
22,25,27	4	0	15
42, 43	24	15	0

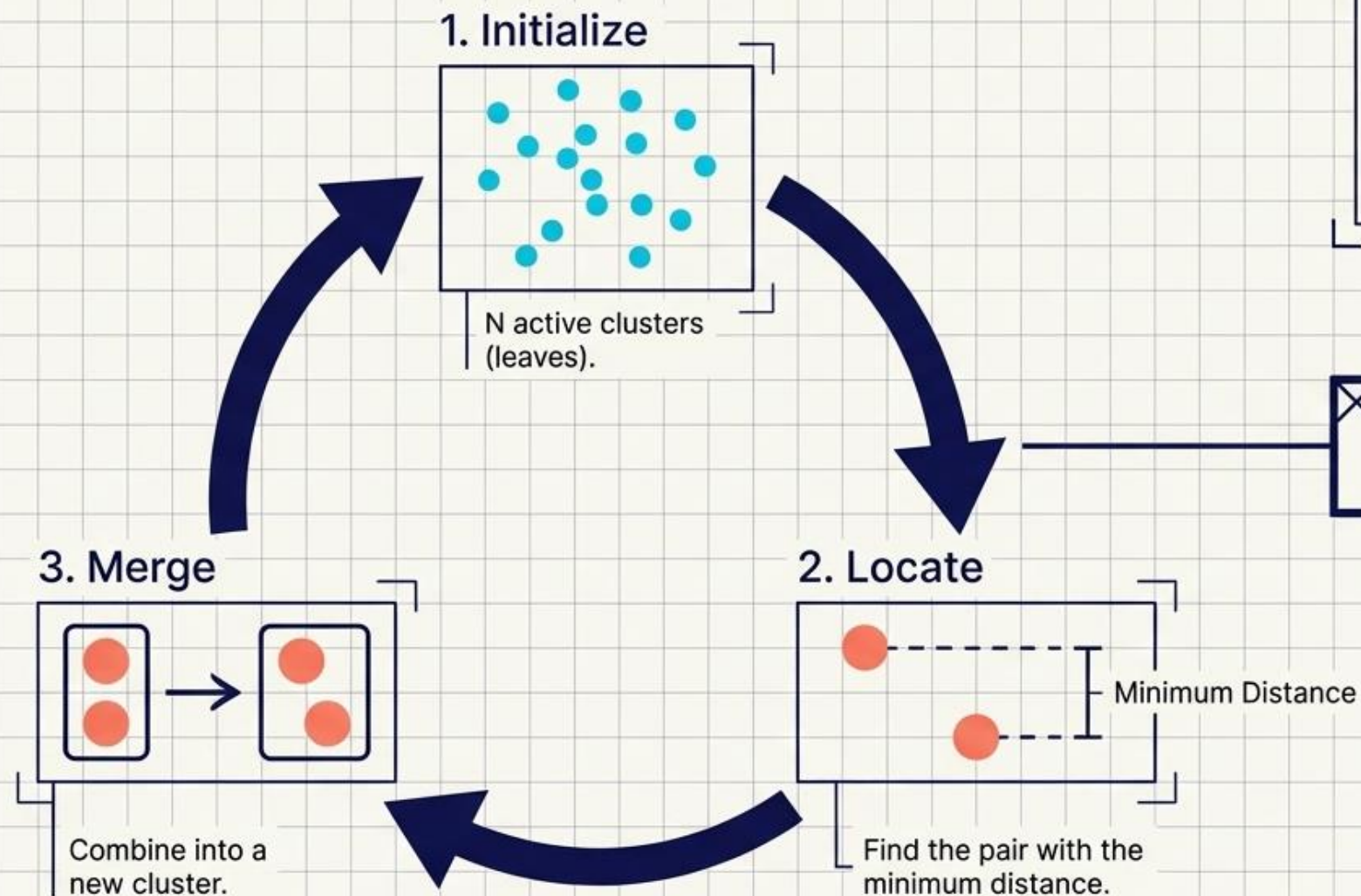


Step 5

	18,22,25,27	42, 43
18,22,25,27	0	15
42, 43	15	0



The Agglomerative Engine



System Notes

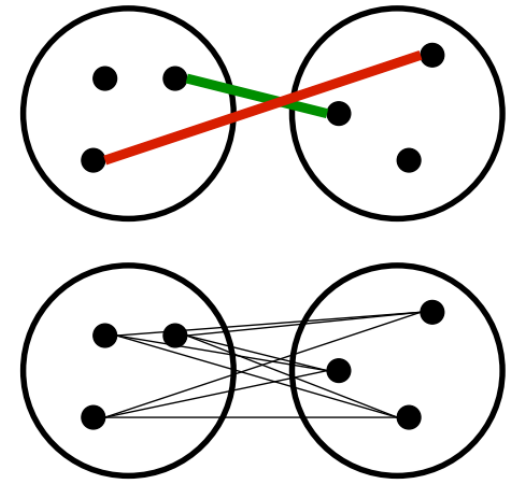
This is a greedy algorithm. It only evaluates the closest neighbors at the current step, building a Minimum Spanning Tree of relationships from the bottom up.

High complexity: $O(n^3)$ time and $O(n^2)$ memory.

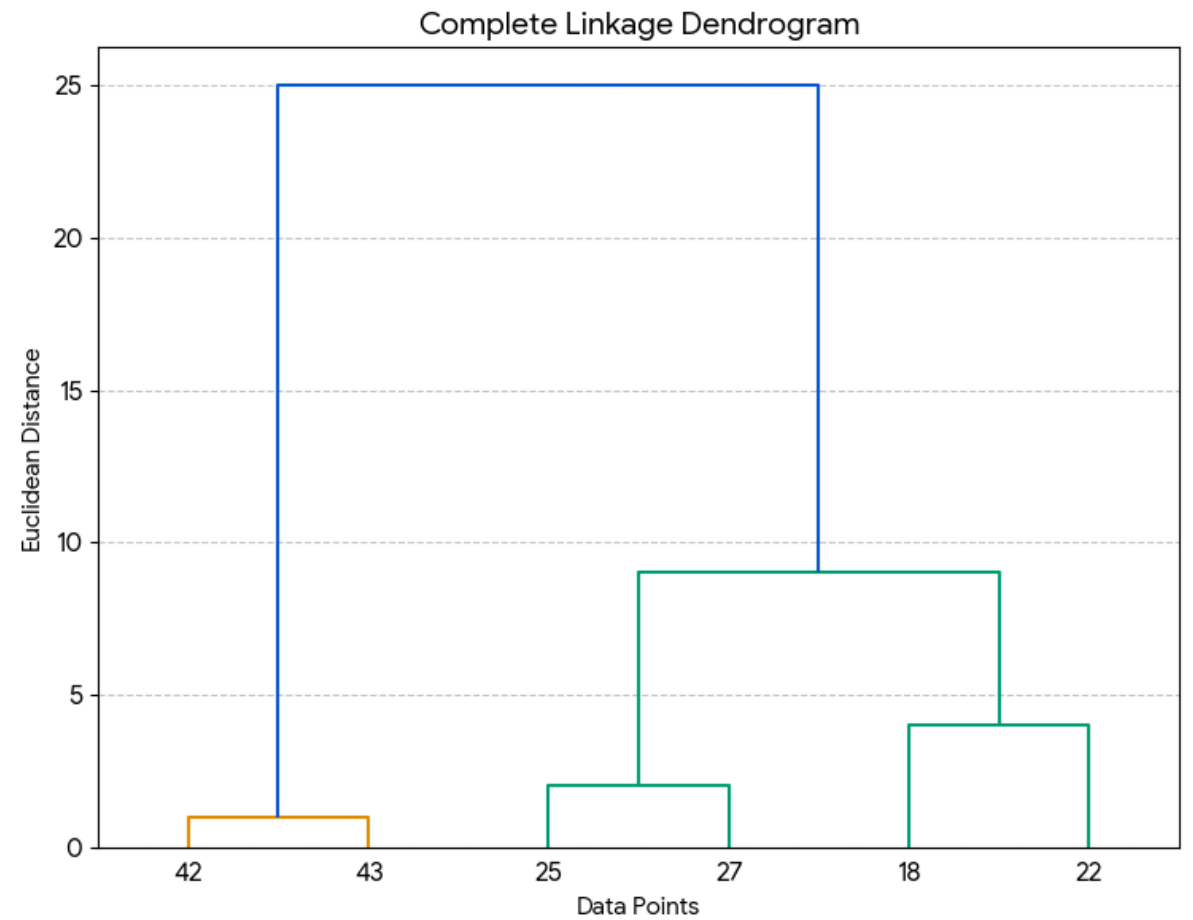
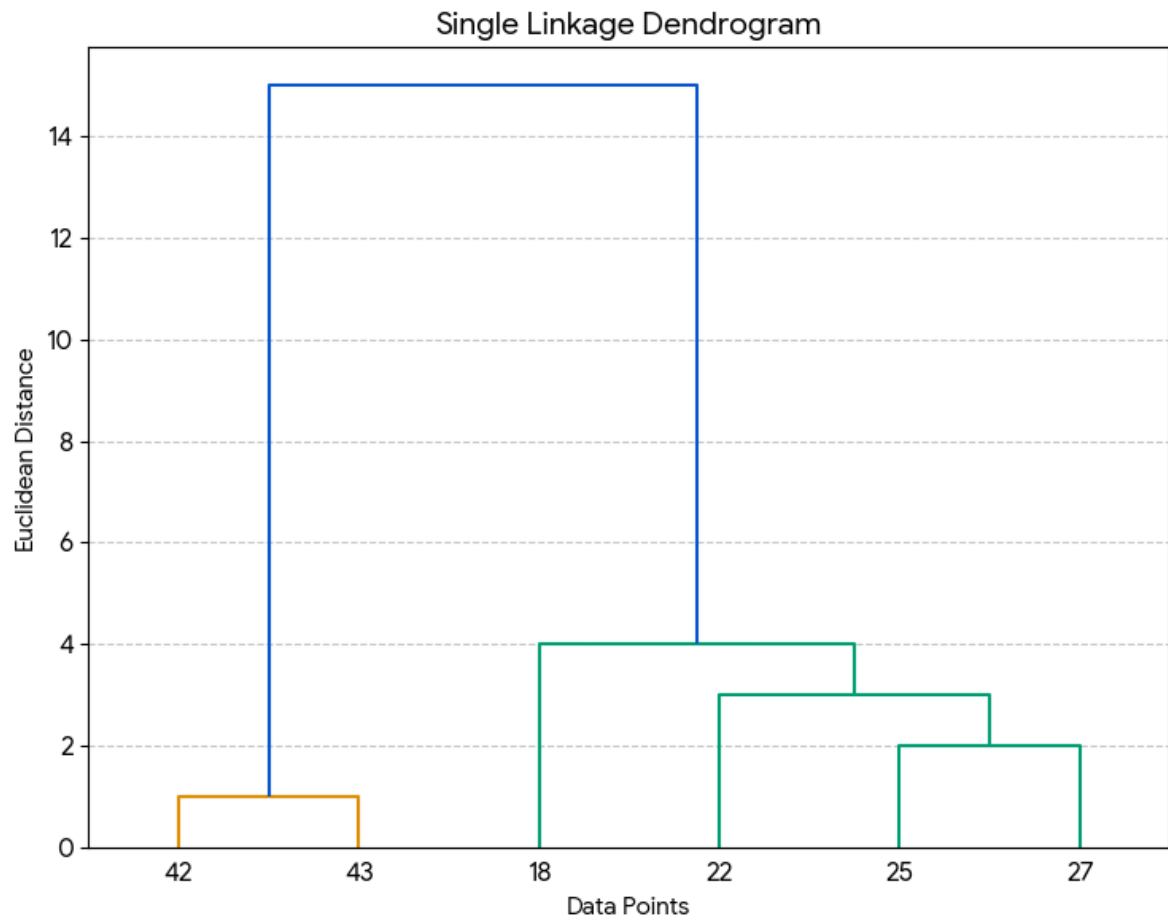
✕ Exit Condition:
Loop ends when
 $|\text{Active Set}| = 1$. ✕

Closest Cluster

- How should we define “closest” for clusters with multiple elements?
- Many options:
 - **Closest pair** (single-link clustering)
 - i.e., previous example
 - **Farthest pair** (complete-link clustering)
 - Average of all pairs
- Different choices create different clustering behaviors



Closest Cluster

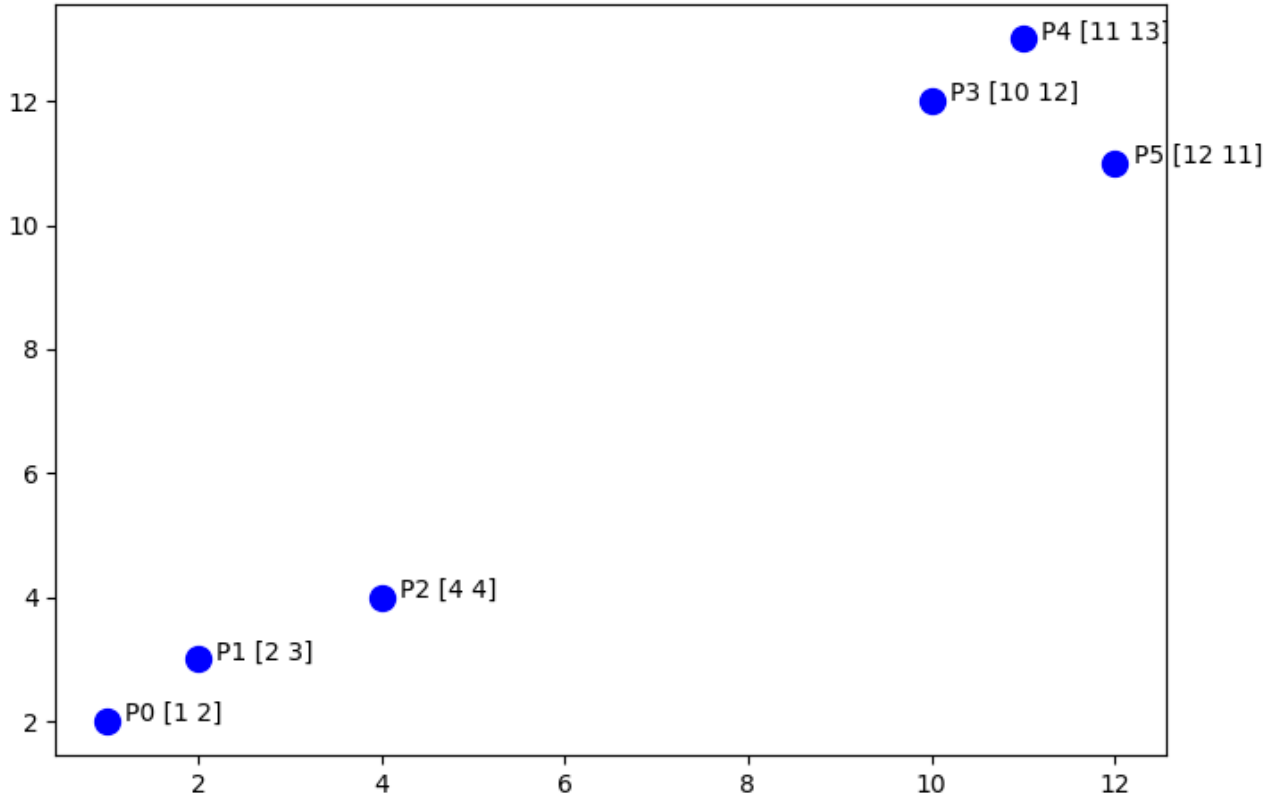


Closest Cluster

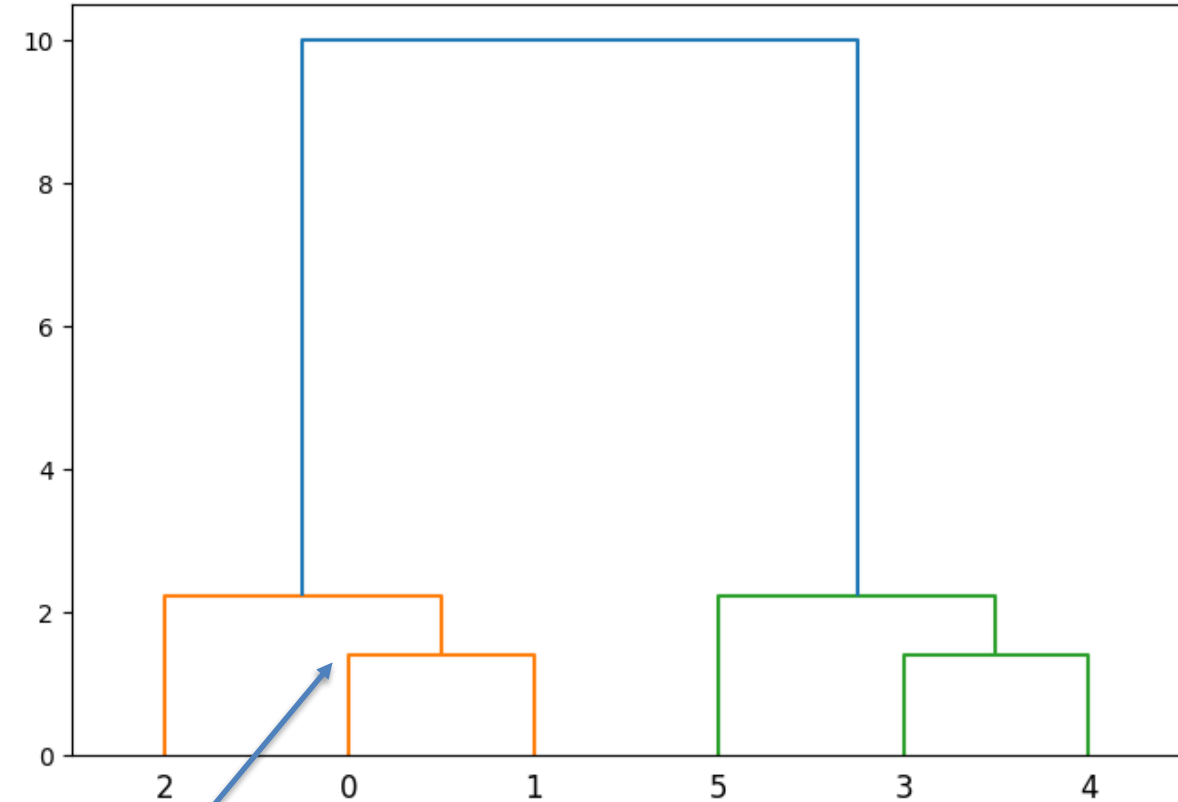
- Single-link:
 - Closest point and closest distance
- Complete Link:
 - Farthest point but closest distance

With 2D data

Original 2D Data Points

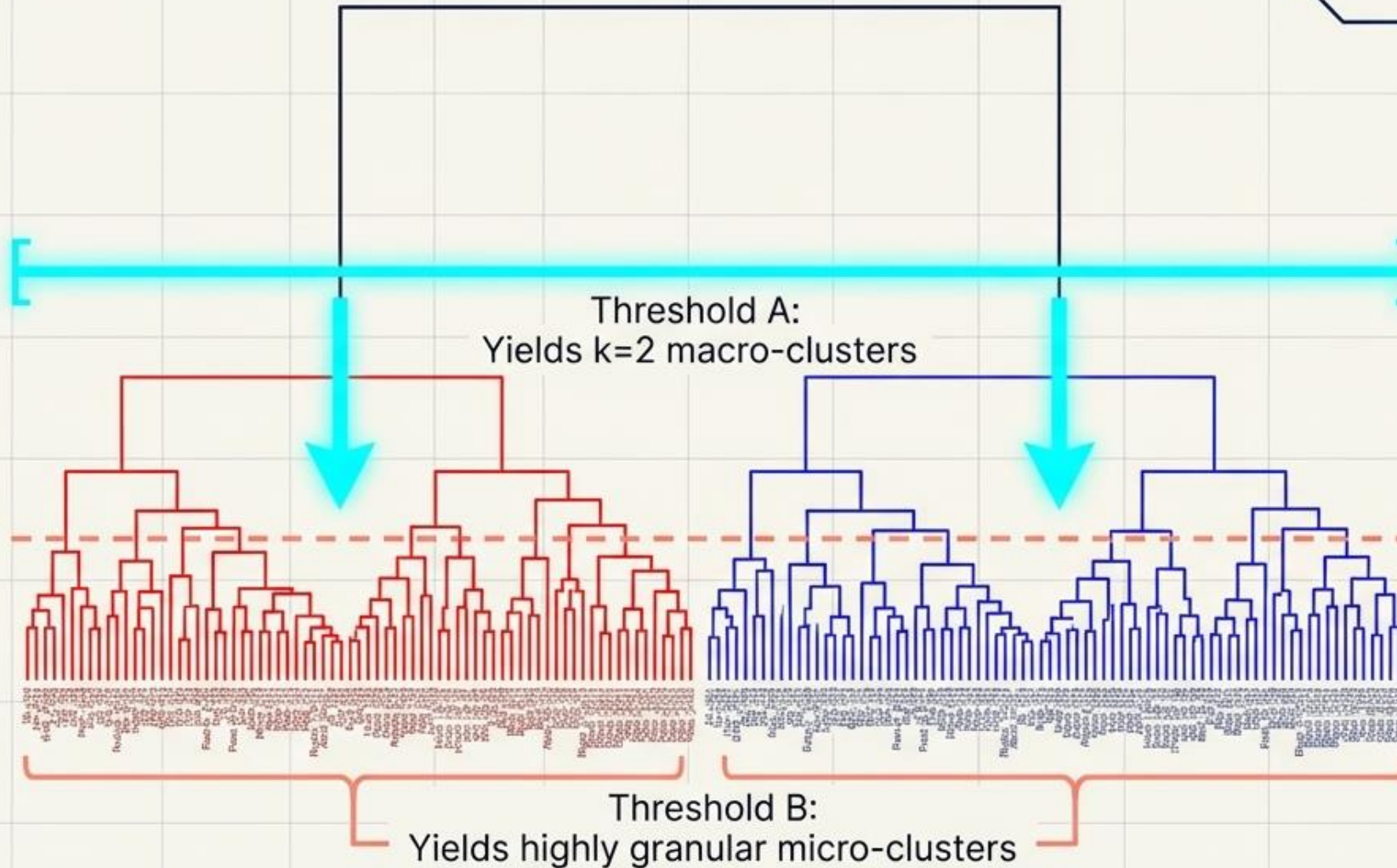


Dendrogram for 2D Data



$$d(P_0, P_1) = \sqrt{(2 - 1)^2 + (3 - 2)^2} = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414$$

Cutting the Tree (Discovering k)



Unlike K-Means, HAC computes **all possible values** of k simultaneously.

You partition the data into flat clusters post-computation by drawing a horizontal threshold.

Lower cut = micro-clusters.

Higher cut = macro-clusters.

Synthesis: Flat vs. Hierarchical Clustering

	K-Means	HAC
Topology	Flat Partitions	Nested Tree
The k Parameter	Must be pre-defined	Emergent; cut post-hoc
Geometric Assumption	Strictly spherical	Highly flexible based on Linkage
Interpretability	Low (black box centroids)	High (exact visual dendrogram)
Computational Cost	Fast ($O(n)$)	Expensive ($O(n^3)$)

Use K-Means for massive datasets and predictive speed. Use Hierarchical Clustering for exploratory data analysis, small-to-medium datasets, and discovering hidden systemic taxonomies.